

1. Likelihood based on record values and times

Your log-likelihood is (rewritten mathematically)

$$\ell(b) = \sum_{i=1}^{m-1} \log f(R_{i+1} - R_i \mid b) + \mathbb{1}_{\{L_m < n\}} \left[-\frac{1}{2} \log(\pi(n - L_m)) \right],$$

where

$$f(x \mid b) = \frac{1}{2b} \mathbb{1}_{\{|x| \leq b\}}, \quad \text{with } \min = -b, \max = b.$$

Only the first term depends on b .

2. Simplify the likelihood

Let

$$D_i = R_{i+1} - R_i, \quad i = 1, \dots, m-1$$

and define

$$M = \max_{1 \leq i \leq m-1} |D_i|.$$

Then

$$\ell(b) = \begin{cases} -(m-1) \log(2b) + C, & b \geq M, \\ -\infty, & b < M, \end{cases}$$

where C collects all terms independent of b .

3. First derivative (score function)

For $b > M$,

$$\boxed{\frac{\partial \ell(b)}{\partial b} = -\frac{m-1}{b} < 0}$$

For $b < M$, the likelihood is $-\infty$ and **the derivative does not exist**.

✔ **Conclusion:**

There is **no interior critical point**.

4. Second derivative

For $b > M$,

$$\frac{\partial^2 \ell(b)}{\partial b^2} = \frac{m-1}{b^2} > 0$$

This confirms the log-likelihood is **convex**, so the maximum occurs on the boundary.

5. Maximum likelihood estimator

Since the likelihood is decreasing for $b \geq M$,

$$\hat{b} = \max_{1 \leq i \leq m-1} |R_{i+1} - R_i|$$

This is a **boundary MLE** (non-regular).

6. Distribution of the estimator (record case)

Under the model,

$$|D_i| \sim \text{Unif}(0, b) \quad \text{i.i.d.}$$

Hence

$$\frac{\hat{b}}{b} = \max_i \frac{|D_i|}{b} \sim \text{Beta}(m-1, 1).$$

Therefore,

$$\mathbb{P}(\hat{b} \leq x) = \left(\frac{x}{b}\right)^{m-1}, \quad 0 \leq x \leq b.$$

7. Mean and variance of the estimator

Mean

$$\mathbb{E}[\hat{b}] = \frac{m-1}{m} b \quad (\text{downward biased})$$

Variance

$$\text{Var}(\hat{b}) = \frac{m-1}{m^2(m+1)} b^2$$

Bias-corrected estimator:

$$\tilde{b} = \frac{m}{m-1} \hat{b}, \quad \text{Var}(\tilde{b}) = \frac{b^2}{(m-1)(m+1)}.$$

8. Why Fisher information does not apply

- The MLE lies on the **boundary**
- The score is never zero
- The second derivative is positive (convex likelihood)

✗ Wald intervals

✗ Score tests

✗ Standard asymptotic normality

✓ Exact order-statistic inference is required

